

Agentic AI Between Capability and Reliability: A Structured Review of Architectures, Design Patterns, Evaluation Practice, and Enterprise Deployment

A Review Article

Indrajit Sarkar

Independent Researcher

Indusarkar01@gmail.com

ORCID:

<https://orcid.org/0009-0000-3038-6379>

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Abstract

Autonomous software agents built on large language models have moved from research demonstrations to production systems in under three years. This review organises the evidence behind that shift and argues that the field's binding constraint has changed. Capability is no longer the scarce resource; reliability under composition and governance under autonomy are. Drawing on peer-reviewed papers, preprints, standards documents, vendor engineering material, and analyst forecasts published between 2022 and mid-2026, we assemble a reference architecture for agentic systems, catalogue the design patterns that have stabilised around it, and trace how memory, planning, orchestration, and tool-invocation subsystems interact. We then confront a discrepancy that runs through the empirical record. On curated benchmarks, agents now approach or exceed human baselines: reported scores on SWE-bench Verified rose from roughly one fifth to four fifths of tasks solved in under a year, and operating-system interaction scores climbed from 12 per cent to above 72 per cent. On live interfaces and extended horizons the same systems complete between 7 and 50 per cent of tasks, and accuracy on long-horizon analysis falls by close to 47 points between early and late turns. We show that a single-parameter compounding model reproduces this gap without appealing to any additional failure mechanism, which reframes long-horizon fragility as an expected consequence of sequential dependence rather than an anomaly. Three further findings follow. Interoperability has converged on two complementary protocols, one for tool access and one for inter-agent messaging, yet neither encodes governance primitives such as authorisation, dissent, or deliberation. Commercial expectations are large and internally inconsistent, with forecasts of roughly one trillion United States dollars in annual agentic software spending by 2030 sitting alongside survey evidence that under a tenth of enterprises operate agents at scale and that two fifths of programmes are expected to be cancelled. Occupational effects to date resemble redefinition rather than substitution, with only one organisation in five reporting reduced service-desk headcount. We propose a Capability-Reliability-Governance framework for reading the literature, a minimum reporting standard for agent evaluation, and six falsifiable propositions for the research agenda through 2030.

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Keywords: Agentic AI; large language model agents; multi-agent systems; agent evaluation; long-horizon reliability; tool use; Model Context Protocol; AI governance; IT service management; workforce transformation

1. Introduction

1.1 Motivation

Software that pursues goals without step-by-step instruction is not a new ambition, but until recently it was not a practical one. That changed quickly. Between 2023 and 2026, systems built by wrapping a language model in a control loop, giving it memory, and connecting it to external tools progressed from laboratory curiosities to components inside commercial products used at industrial volume. One vendor reported 3.2 million users, 13 million coding sessions, and 761 million model invocations in a single month of production telemetry [1]. Whatever one concludes about the durability of the trend, the volume is no longer speculative.

The distinguishing property of these systems is not fluency. It is persistence. A conventional language model answers a prompt and forgets it. An agent holds an objective across many turns, decides what to do next, acts on the world through an interface, observes what happened, and revises. Autonomy of this kind is what practitioners now label *agentic*, and it changes the engineering problem in a specific way: errors stop being isolated bad answers and become states that later steps inherit.

Commercial expectations have scaled accordingly. One analyst house projects annual value creation between 2.6 and 4.4 trillion United States dollars across more than sixty enterprise applications [2]. Another forecasts 1.15 billion active agents by 2029, collectively performing on the order of 217 billion actions each day [3]. A third puts agentic software spending near 985 billion dollars by 2030 [4], and expects two out of five enterprise applications to embed task-specific agents by the end of 2026 [5].

1.2 The gap this review addresses

Existing surveys are numerous and, individually, competent. They tend to be narrow by design: one treats planning topologies, another multi-agent coordination, a third benchmark construction, a fourth security. What the literature lacks is a synthesis that carries an argument across those boundaries and connects the technical record to the deployment record. That connection matters because the two records disagree. Benchmark curves point upward steeply. Enterprise outcomes do not.

Reviews also tend to inherit the promotional register of the material they summarise. Analyst forecasts are quoted as findings, vendor claims as measurements, and preprints as settled results. We take a different stance. Where evidence is weak we say so, we grade sources explicitly (Section 2.3), and we treat inconsistency between forecast and observation as a phenomenon worth explaining rather than a blemish to smooth over.

1.3 Contributions

This review makes five contributions.

A reference architecture with its patterns attached. Section 5 consolidates the subsystem decompositions proposed across the literature into one description of the agent control loop, and maps the canonical design patterns onto positions within it.

A quantitative account of the benchmark-to-production gap. Section 9.3 shows that a single-parameter compounding model, with no additional assumptions about hallucination or environmental drift, reproduces the observed collapse in long-horizon performance. This reframes the reliability problem: the issue is less that models reason poorly than that sequential composition is unforgiving of any per-step error rate above a very small threshold.

A minimum reporting standard for agent evaluation. Section 9.4 proposes five quantities that any agent evaluation should report together, on the grounds that single-attempt success rates systematically overstate deployability.

The Capability-Reliability-Governance framework. Section 14 argues that progress along three axes has been badly uneven, that the neglected axes now dominate deployment failure, and that research effort is misallocated relative to where value is lost.

Six falsifiable propositions. Rather than closing with an open-ended wish list, Section 14.2 states predictions specific enough to be wrong.

1.4 Organisation

Section 2 documents method. Section 3 places agentic systems in a longer lineage. Section 4 fixes definitions and separates agentic AI from neighbouring technologies. Sections 5 through 8 treat architecture, memory, planning, and multi-agent organisation. Section 9 covers evaluation and develops the compounding argument. Section 10 addresses security, privacy, alignment, and regulation. Sections 11 through 13 turn to deployment economics, regional trajectories, and effects on service work. Section 14 synthesises. Section 15 states the limits of the review itself, and Section 16 concludes.

2. Review Method

2.1 Corpus construction

We assembled material published between January 2022 and July 2026, with earlier work admitted where it is foundational (the Transformer architecture, for instance, or early formal treatments of partially observable decision processes). Four source classes were searched: archival venues in machine learning and software engineering; the arXiv preprint server; engineering documentation and technical blogs maintained by model providers and framework maintainers; and market research from established analyst firms.

Queries combined a paradigm term (*agentic AI, LLM agent, autonomous agent, multi-agent LLM*) with a topic term (*architecture, memory, planning, tool use, benchmark, reliability, governance, service desk, workforce*). Reference chaining from highly cited entries supplemented keyword search, as did forward citation tracking on the seven papers that recur most often across independent surveys.

2.2 Screening

Items were retained when they satisfied at least one of four conditions: they introduced an architecture, pattern, or protocol; they reported quantitative results on a recognised benchmark; they documented a deployment with measurable outcomes; or they provided survey or forecast data on adoption and labour effects. We excluded opinion pieces without underlying data, marketing material that reported no method, and duplicate reporting of a single primary result across secondary outlets. Where several outlets reported one study, we cite the primary source.

2.3 Evidence grading

Reviews in this area routinely place a peer-reviewed benchmark result and a vendor press release side by side without comment. We instead assign each cited claim to one of three tiers, and we use tier language in the text so readers can discount appropriately.

Table 1. *Evidence tiers used throughout this review.*

Tier	Description	Treatment in this review
A	Peer-reviewed results, or preprints with released code, data, and reproducible protocol	Reported as findings
B	Vendor engineering documentation, production telemetry, standards texts, and preprints without artefacts	Reported with attribution and hedged verbs
C	Analyst forecasts, survey summaries, single-organisation case studies, encyclopaedic sources	Reported as expectations or indications, never as measurement

Three consequences follow. Market figures in Sections 11 and 12 are Tier C throughout and are discussed as sentiment indicators rather than as predictions with error bars.

Benchmark numbers are Tier A or B, and we name the evaluation protocol wherever it is available, because agent scores are protocol-sensitive to a degree that raw percentages conceal. Several claims present in the underlying draft rested on a single encyclopaedic source and could not be traced to a primary document; we removed those claims rather than launder them through a citation.

2.4 Synthesis approach

Evidence was organised by subsystem rather than by publication, so that competing proposals for, say, memory management appear together and can be compared. Numerical claims were tabulated with their evaluation conditions attached. Where two sources conflict, both figures appear and the discrepancy is discussed. Section 9.3 departs from summary and offers an original quantitative argument; it is flagged as such.

3. From Symbolic Systems to Autonomous Agents

The intellectual line running to present-day agents is roughly seven decades long, and each phase contributed something still visible in current designs.

Symbolic AI, dominant from the mid-1950s into the 1970s, treated intelligence as the manipulation of explicit structures. The General Problem Solver and the Lisp language established the working assumption that reasoning could be represented, inspected, and searched. Expert systems extended the idea in the 1970s and 1980s by pairing curated knowledge bases with inference engines. DENDRAL, MYCIN, and R1/XCON worked well inside their boundaries and failed abruptly outside them. That failure mode is instructive, because contemporary agents inherit a version of it: competence remains bounded, and the boundary is not visible from inside the system.

Statistical learning displaced hand-coded knowledge through the 1980s and 1990s. Rather than encoding what to conclude, systems induced it from data using backpropagation, kernel methods, and ensembles. Deep learning then scaled representation learning through the 2000s and 2010s, from belief networks to convolutional architectures, culminating in the Transformer, whose attention mechanism made sequence modelling parallelisable and therefore scalable.

Large pretrained models followed. By 2020 it was clear that scale produced behaviours nobody had trained for directly, including in-context learning and instruction following [6]. But these systems remained reactive. They generate on demand, hold no objective between calls, and possess no representation of an environment they might act upon [7].

Three developments converted reactive models into agents. First, interleaving reasoning traces with actions in a thought-action-observation cycle produced behaviour that was both more capable and easier to inspect than either pure reasoning or pure action selection [8]. Second, verbal self-critique stored in episodic memory allowed improvement across attempts without touching model weights [9]. Third, a widely adopted architectural summary organised agent design around planning, memory, and tool use, giving practitioners a shared vocabulary [8]. Systems including AutoGPT, Voyager, and simulated societies of generative agents then demonstrated sustained operation over long horizons [8], [10].

A more recent framing describes the current transition as one from pipeline architectures, where orchestration logic sits outside the model and sequences its calls, toward model-native designs in which planning and tool selection are internalised in the weights [11]. Reinforcement learning supplies the training signal for that internalisation [11]. The distinction is not merely implementational. Pipeline systems expose their control flow, which makes them auditable; model-native systems hide it, which makes them harder to govern. Section 10.4 returns to this trade-off.

4. Defining the Object of Study

4.1 Definitions in circulation

No canonical definition has been ratified, and the working ones emphasise different properties. One academic treatment frames the paradigm as a qualitative jump in reasoning and interaction capacity sufficient to sustain autonomous behaviour on complex tasks [12]. Another takes a component view, identifying multi-step reasoning, persistent memory, inter-agent interaction, and purposeful tool use as the constitutive parts [13]. A management-facing account foregrounds behaviour instead: independent decision-making, action, and adaptation without continuous supervision [14]. Industry definitions add degrees, describing action automation at varying levels of autonomy [15], or simply describe programs that understand, plan, and execute [16].

The variation is not merely terminological sloppiness. Component definitions invite checklist compliance, which vendors can satisfy trivially. Behavioural definitions resist that but are hard to test. For the purposes of this review we adopt a minimal operational criterion: a system is agentic when the sequence and number of its model invocations are determined by the system itself, in response to observations, rather than fixed in advance by its caller. This criterion is decidable from a system's execution trace, which matters for the evaluation argument in Section 9.

4.2 Agent and agentic

A useful distinction separates the individual agent from the paradigm. Individual agents are modular, model-driven components aimed at specific automation tasks. Agentic AI names the systemic arrangement: several agents collaborating, tasks decomposed dynamically, memory persisting across episodes, and autonomy coordinated rather than merely delegated [7]. Confusing the two produces inflated claims, since much of what is marketed as agentic AI is a single agent with a tool list.

4.3 Autonomy scales

Several stratifications compete. A four-level industrial hierarchy grades systems by decision complexity [17]. A five-level scheme indexes autonomy to the shrinking human role, moving through operator, collaborator, consultant, approver, and observer [18]. The most granular proposal spans seven tiers, from stateless models to formally verified hierarchical systems, and decomposes each tier along perception, memory, planning, tool use, and collaboration [19].

We find the human-role formulation the most useful of the three, for a reason that becomes important in Section 13: it makes the human position explicit, and it is the human position, not the model capability, that determines whether a deployment is auditable.

4.4 Differentiation from adjacent technologies

Agentic systems are frequently conflated with four neighbours: conversational interfaces, robotic process automation, bare language models, and assistant products. Table 2 separates them along the dimensions that matter operationally.

Table 2. Operational comparison of agentic systems against adjacent technologies. Compiled from [7], [20], [21].

Dimension	Chatbots	RPA	Bare LLMs	AI assistants	Single agents	Agentic systems
Autonomy	Low	Low	Low to moderate	Moderate	High	High and coordinated
Planning	None	None	Single pass	Rudimentary	Multi-step	Dynamic decomposition
Memory	Session only	None	Context window	Cross-session	Short and long term	Persistent with retrieval
Tool use	Minimal	Interface scripting	None unless augmented	Constrained	API invocation	Multi-tool orchestration
Reasoning	Pattern matching	Conditional rules	Single-pass generation	Contextual	Interleaved reason-act	Deliberation with reflection
Multi-agent	None	None	None	None	Limited	Constitutive
Adaptability	Static	Breaks on interface change	Prompt-dependent	Moderate	Feedback-driven	Adapts to novel states
Failure handling	Escalates	Halts on exception	May fabricate	Requests clarification	Self-correction	Recovery with causal tracing

Two entries deserve caution. The self-correction and recovery columns describe design intent more than measured behaviour; Section 7.3 reports evidence that reflection often recycles the original error. And the adaptability column understates how brittle agents remain when an interface changes underneath them, a failure mode they share with the robotic process automation they are said to supersede.

5. Reference Architecture and Design Patterns

5.1 The control loop

At runtime an agentic system repeats a cycle that constitutes its basic unit of behaviour. It assembles context, invokes the model, inspects the output for actionable directives, executes any tool calls, delegates to specialists where appropriate, and returns a consolidated result only when no work remains outstanding [22]. Everything else in the architecture exists to serve, constrain, or observe that loop.

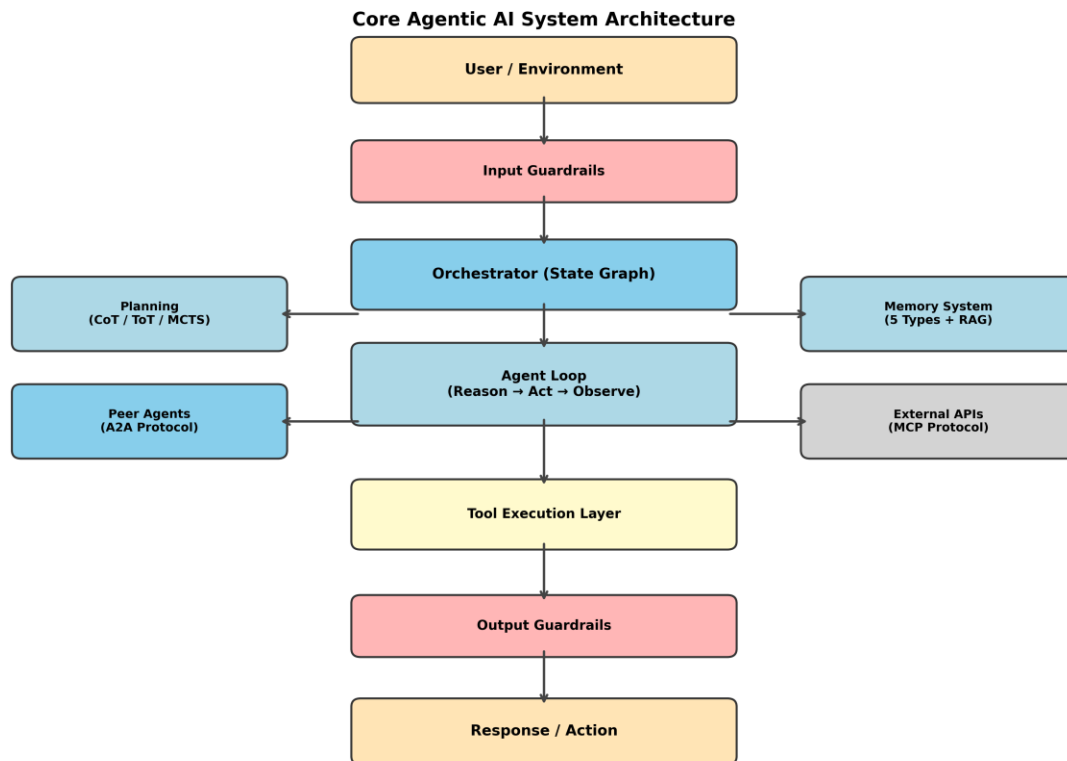


Figure 1. Reference architecture for an agentic system. Vertical flow runs from input through guardrails, orchestration, the agent loop, tool execution, and output validation. Planning and memory subsystems attach laterally to the loop; peer agents and external services attach through interoperability protocols.

5.2 Subsystem decomposition

Two decompositions in the literature are worth reconciling. A system-theoretic account identifies five interacting subsystems: reasoning and world modelling, perception and grounding, action execution, learning and adaptation, and inter-agent communication [23]. A component-level taxonomy instead enumerates the policy model, memory modules, world models, planners, tool routers, and critics, then treats orchestration separately

across single-agent and multi-agent configurations under centralised or decentralised control [24].

These are not competitors. The first describes functions, the second describes parts, and a given part may implement several functions. We adopt the functional view when discussing behaviour and the component view when discussing implementation, and we note that no published system implements all five functions with equal maturity. Learning and adaptation is the weakest in practice: most deployed agents do not improve from their own operational history at all, a point Section 14 treats as a research priority rather than an engineering detail.

5.3 Canonical patterns

Four patterns recur across implementations, and practitioner accounts have converged on essentially the same set [25].

Interleaved reasoning and action. The pattern alternates verbal reasoning with environmental action in a loop that continues until the task terminates [26]. On interactive benchmarks it improved absolute success rates over imitation and reinforcement learning baselines by 34 and 10 points respectively [27]. Its weakness is structural: it is greedy, committing to each action without lookahead.

Reflection. After a failed attempt, the agent writes a critique of its own trajectory in natural language, stores it, and consults it on the next attempt [28]. No gradients are involved, which makes the pattern cheap and portable.

Search-augmented planning. Monte Carlo tree search over reasoning and action sequences, with the model serving simultaneously as policy, value estimator, and critic, reported 94.4 per cent on a code synthesis benchmark [29]. The cost is severe, on the order of dozens of model calls per task.

Plan then execute. Strategic decomposition is separated from tactical execution so that a strong planner produces an abstract plan and a cheaper executor implements each step [30], [31]. The separation buys cost control and interpretability at the price of replanning latency when execution diverges from the plan, and recent variants schedule replanning adaptively rather than on failure alone [32].

The four are complements rather than alternatives. Production systems typically compose them: plan-and-execute at the outer level, interleaved reasoning within each step, reflection on failure, and search reserved for the small subset of decisions that justify its cost. We are not aware of published work that characterises the optimal composition, which is a concrete and tractable open problem.

5.4 Formal framing

The decision problem admits a standard formulation as a Markov decision process $M = (S, A, T, R, \gamma)$ in which the agent maximises expected discounted return

$$J(\pi) = E[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)] \quad (1)$$

where S is the state space, A the action space, T the transition kernel, R the reward function, and γ the discount factor [33]. Real deployments violate the full-observability assumption, so the partially observable extension is the more faithful model [34]. Orchestration frameworks add a second formalism, representing workflows as directed graphs $G = (V, E, \delta)$ with node set V , control and data edges $E \subseteq V \times V$, and a state-parameterised transition function $\delta: V \times S \rightarrow V$ [35].

The formalisms are less useful than they look, and it is worth saying why. Neither the reward function nor the transition kernel is available in the settings where agents are deployed. What the formalism does provide is a vocabulary for stating precisely which quantity a given method estimates, and Section 9.3 exploits exactly that precision.

5.5 Boundary enforcement

Safety in production is implemented as layered filtering rather than as a property of the model. One widely used framework distinguishes input rails that sanitise requests before invocation, retrieval rails that filter knowledge brought into context, and output rails that validate generations before release [36]. A complementary design separates automated checks from human approval, defining explicit conditions under which execution proceeds, pauses for review, or halts [37]. The second distinction is the more consequential one, because it locates authority rather than merely filtering content.

5.6 Interoperability, and what the protocols omit

Two protocols have become de facto standards along orthogonal axes. A tool-access protocol introduced in November 2024 specifies a client-server interface for typed data exchange and tool invocation between applications and external services [38], [39]. An inter-agent protocol released in April 2025 defines semantics for delegation and cooperation between agents [40]. Together they cover vertical integration to tools and horizontal integration to peers [41]. Governance of both moved to a neutral foundation during 2025 and 2026 [42].

Convergence on transport is genuine progress. But an analysis of the protocol family finds that governance primitives are absent across all of them: none specifies voting, deliberation, or the preservation of dissent when agents disagree [43]. This is a substantive omission rather than a missing feature. A multi-agent system without a decision procedure has no principled way to resolve conflict, so conflict resolves by whichever agent writes last. Section 14.3 sketches of what a governance layer above these protocols would need to carry.

6. Memory and Retrieval

6.1 Taxonomy

Agentic memory is organised in tiers that borrow their vocabulary, though not their mechanisms, from cognitive psychology. The foundational split is between short-term memory realised as in-context information inside a finite window, and long-term memory realised as external storage with a retrieval interface [8]. Framework documentation extends this with procedural, semantic, and episodic categories, and separates thread-scoped memory for an active conversation from cross-session memory namespaced per user [44]. A 2025 survey advances the taxonomy further, distinguishing parametric from contextual representation and identifying six atomic operations: consolidation, updating, indexing, forgetting, retrieval, and compression [45].

Forgetting deserves emphasis, since it is the operation most often omitted from implementations. A system that never discards accumulates contradictions, and contradiction in a retrieval store is worse than absence because it is retrieved with equal confidence.

Table 3. *Memory tiers in agentic systems and their implementations.*

Memory type	Content	Storage mechanism	Temporal scope	Representative systems
Working	Information for the current step	Context window, message buffer	Single session	Graph and conversation orchestrators [46]
Long-term	Knowledge persisted across sessions	Vector indices	Cross-session	Virtual-context and crew frameworks [47], [48]
Episodic	Records of specific past events	Memory streams, episodic buffers	Event-specific	Simulated societies, reflective agents [49], [9]
Semantic	Entities and their relations	Knowledge graphs, entity stores	Atemporal	Graph-structured retrieval systems [50]
Procedural	Learned skills and reusable code	Skill libraries	Persistent	Open-ended embodied agents [10]

6.2 Systems that treat memory as first class

Three lines of work push past retrieval-as-lookup. One reconceives memory management as a learned skill by placing read, write, search, and append operations in the same action space as task actions, so the agent decides what to remember as part of deciding what to do [51]. A second elevates memory to a managed operating resource through a unified abstraction spanning parametric, activation, and plaintext modalities, with explicit organisation and governance interfaces [52]. A third maintains complete experience

records as natural language streams, periodically synthesises them into higher-level reflections, and retrieves relevantly when planning [49].

The common insight is that retrieval quality is a policy problem, not an indexing problem. Which memory to consult is a decision, and decisions can be trained.

6.3 Agentic retrieval

Classical retrieval-augmented generation runs a fixed pipeline: receive query, search, assemble, generate. Agentic retrieval restructures this by treating search as one tool among several, so the system reasons about what it needs, selects a strategy, judges whether what came back is sufficient, and iterates [53]. Four familiar patterns, reflection, planning, tool use, and multi-agent collaboration, are recruited to manage retrieval dynamically in response to signals about information quality [54]. A systematisation paper supplies formal grounding by modelling the retrieval-generation loop as a finite-horizon partially observable decision process with explicit policies and state transitions [55].

Specific mechanisms have proved effective. Training a model to decide when to retrieve, using dedicated reflection tokens, outperformed strong baselines across task categories [56]. Adding an explicit relevance evaluator that filters retrieved documents functions as a plug-in corrective layer [57]. Constructing a knowledge graph from a corpus before querying it improves multi-hop and summarisation performance through systematic entity and relationship extraction [50].

6.4 Context accumulation over long horizons

Extended sequences expose a scaling problem sometimes called context bloat. As history accumulates, cost grows faster than linearly, latency rises, and answer quality degrades because attention is diluted across material that includes the agent's own earlier mistakes [58]. The last clause is the important one. Recorded failures are retrieved alongside successes and can be mistaken for precedent.

Two approaches address the constraint. Virtual context management, modelled on operating-system paging, moves information between an in-context working set and external storage to simulate unbounded memory [47]. A learned controller that manages long-term and short-term stores jointly reported consistent gains over memory-augmented baselines across five long-horizon evaluations [59]. Neither eliminates the problem; both shift where the loss occurs.

7. Planning, Reasoning, and Self-Correction

7.1 Reasoning topologies

The structure of intermediate computation has evolved from a line to a graph. Chain-of-thought prompting established that decomposing a problem into explicit intermediate steps improves arithmetic, commonsense, and symbolic performance, with benefits concentrated in models above roughly one hundred billion parameters [60]. A subsequent analysis carries an uncomfortable implication: relevance and ordering of steps matter more than their validity, and demonstrations containing invalid reasoning still recover 80 to 90 per cent of full performance [61]. Whatever chains of thought are doing, they are not straightforwardly proof.

Tree-structured search generalises the chain by branching at decision points and permitting backtracking, at substantially higher inference cost [62], [63]. Graph-structured reasoning generalises further to arbitrary directed acyclic structures in which reasoning units are vertices and dependencies are edges, which permits aggregation and refinement operations that a tree cannot express [64]. A unifying taxonomy analyses how these structural choices trade performance against compute [65].

Table 4. Reasoning and planning approaches compared.

Approach	Structure	Relative compute	Principal strength	Principal weakness
Chain of thought	Linear	1×	Simple; emerges with scale	No backtracking; errors propagate [60]
Tree of thoughts	Tree with search	Exponential in depth	Explores alternatives; backtracks	Branching cost grows exponentially [62]
Graph of thoughts	Directed acyclic graph	Variable, high	Supports merging and refinement	Orchestration complexity [64]
Interleaved reason-act	Loop	Moderate	Environmentally grounded; inspectable	Greedy; no lookahead [66]
Reflection	Multi-trial episodic loop	Moderate	Learns from failure without retraining	May recycle the original error [67], [68]
Tree search with reflection	Search plus critique	Very high, dozens of calls	Unifies reasoning, acting, planning	Expensive; needs value estimates [69]

7.2 Search-based planning

Monte Carlo tree search transferred from game playing to language-model planning with reasonable success. One method casts the model as both world model and reasoning agent, using tree search to explore solution space [70]. A deeper integration unifies reasoning, acting, and planning with model-derived value functions and self-reflective feedback [69]. Comparative work on short-horizon tasks found tree search achieved the best solution quality among planners tested, while bidirectional methods were more efficient overall [71]. A graph-augmented variant reported complete success on synthetic planning tasks against 92 per cent for tree search with reflection and 64 per cent for the interleaved pattern, and required no model calls during planning against 37 for the comparison method [72]. That last figure, if it replicates, matters more than the accuracy difference, since planning cost is what keeps search out of production.

7.3 Reflection and process supervision

Reflection equips an agent with dynamic memory and verbal self-critique: given feedback, it writes an analysis, stores it, and conditions later attempts on it, with no parameter updates [67], [9]. A code-specific variant has the model explain its own program in natural language while inspecting execution output, improving baseline accuracy by up to 12 points [73].

The mechanism has a documented failure mode that tempers enthusiasm. On difficult instances, self-reflections tend to restate the original misconception rather than open a new line of reasoning [68]. This is what one would expect if reflection samples from the same distribution that produced the error, and it suggests that useful critique requires either an independent critic or external evidence.

Process reward models address a related problem by supervising intermediate steps rather than only final answers, using sampling to estimate step-level correctness, and they outperform outcome-only evaluation. We were unable to trace the specific quantitative claim made for this method in the source material to a primary publication, so we report the direction of the result without a figure.

7.4 Limits

Autonomous planning by language models remains weak in absolute terms, and the gap between benchmark reasoning and planning competence is one of the most robust findings in the literature. A frontier model of 2023 achieved roughly 12 per cent average success across standard planning domains [74]. A reasoning-specialised model reached only 15.6 per cent on complex realistic planning tasks, producing solutions that were valid but redundant [75], [76]. One theoretical account attributes such failures to static planning procedures that cannot detect the boundaries of their own reasoning capacity, formalised as a non-autonomous stochastic dynamical system [77].

Self-verification is the specific weak point. Verification performance collapses under adversarial conditions, while sound external verification produces large gains [78]. The practical reading is direct: hybrid designs that pair generative reasoning with an

independent verifier are the credible route to reliable planning, and architectures that ask a model to check itself should be treated as unverified by default.

8. Multi-Agent Organisation and Tool Use

8.1 Frameworks

Five frameworks anchor the multi-agent ecosystem, each encoding a different theory of how work should be divided.

A role-playing framework established inception prompting to sustain cooperation between two agents without human intervention. An assembly-line framework encodes standardised operating procedures as prompt sequences and distributes specialised roles such as product manager, architect, and engineer [79], [80]. A chat-chain framework organises software development through structured sequential conversation with mechanisms aimed at reducing fabrication [81]. A conversable-agent framework supports flexible modes mixing models, humans, and tools, with group chat and an event-driven runtime for dynamic orchestration [82], [83]. A production-oriented framework structures agents into role-based teams for collaboration and event-driven flows for controlled execution, built without dependency on earlier orchestration libraries [84].

Table 5. Multi-agent framework families compared.

Framework family	Organising principle	Coordination	Communication	Primary domain
Role-playing	Paired roles, inception prompting	Peer to peer	Natural language dialogue	Research exploration
Assembly line	Standardised operating procedures	Hierarchical, manager roles	Structured artefacts	Software development [79]
Chat chain	Sequential waterfall phases	Pipeline	Chained conversation with verification	Software development [81]
Conversable agents	Flexible interaction patterns	Dynamic, event-driven group chat	Multi-turn conversation	General purpose [82]
Crews and flows	Role-based teams plus workflows	Hierarchical and event-driven	Task delegation	Production automation

8.2 Topology and the cost-accuracy frontier

A 2026 survey proposes three topologies, centralised, decentralised, and hierarchical, each optionally augmented with dynamic control [85]. One framework selects among parallel, sequential, hierarchical, and hybrid arrangements by analysing the task dependency graph [86].

The most useful evidence here is economic rather than architectural. Benchmarking across ten thousand regulatory filings found that self-correcting architectures achieved the highest field-level F1 score of 0.943 at 2.3 times baseline cost, while hierarchical architectures sat on the favourable part of the Pareto frontier at 0.921 for 1.4 times cost, and hybrids recovered 89 per cent of the accuracy gain at 1.15 times cost [87]. Hierarchical decomposition with a central planner coordinating domain specialists reported 83.39 per cent on a general assistance benchmark [88].

Read together, these results argue against maximal architectures. The marginal accuracy purchased by self-correction is real but expensive, and most of it is available at a fraction of the price. Cost-blind architecture selection is the most common avoidable error we observed in the deployment literature.

8.3 Emergent specialisation and its ceiling

Two 2026 findings sit in productive tension. The first reports that with minimal scaffolding, agents spontaneously develop specialised roles, decline tasks outside their competence, and form shallow hierarchies without assigned roles; self-organising protocols outperformed centralised coordination by 14 per cent, with a 44 per cent quality spread between protocols, scaling sub-linearly to 256 agents and generating over five thousand distinct roles from eight initial agents [89]. The second identifies empirical saturation thresholds beyond which additional agents stop helping, with a predictive model selecting the optimal architecture in 87 per cent of held-out configurations [90]. A third result complicates the scaling narrative from the other direction: small multi-agent systems can outperform substantially larger single agents even when the single agent has direct tool access [91].

The synthesis we draw is that agent count is a poorly chosen design variable. What the saturation result implies is that coordination overhead grows with population while marginal expertise does not, so the useful question is not how many agents but how many distinct competencies the task actually requires.

8.4 Tool use and the surface it creates

Tool augmentation progressed from self-supervised learning of application interfaces [92] to native function calling with structured outputs across all major platforms [93], [94]. Two efficiency ideas stand out: a two-phase arrangement in which a strong model writes reusable utility functions and a light model applies them [95], and explicit tool creation to separate abstract from concrete reasoning [96].

Benchmark results and field results diverge sharply, and this divergence is the empirical core of Section 9. Reinforcement-learning-trained models reach 86 per cent overall accuracy on a standard function-calling leaderboard [97], and a specialised eight-billion-parameter model reports 89 per cent [98]. Against live interfaces with real variability, completion rates fall to between 7 and 47 per cent for standalone models, reaching at best 50 per cent in interactive agent configurations [99].

Security is the unresolved problem rather than a secondary concern. Prompt injection subverts guardrails and triggers unauthorised invocation [100]. Indirect injection through adversarial content embedded in data the agent processes provides a manipulation channel the user never sees [101]. Protocol-level weaknesses include absent capability attestation, bidirectional sampling without origin authentication, and implicit trust propagation across multiple connected servers [102]. Documented mitigations include dual-model architectures that separate privileged from quarantined processing, sandboxed execution, and least-privilege credentials [103].

9. Evaluation

9.1 Benchmark landscape

Agent evaluation matured rapidly from 2023. The first systematic multi-environment suite assessed models as agents across eight settings including operating-system interaction, databases, knowledge graphs, and web browsing, and identified weak long-horizon reasoning as the dominant failure mode [104], [105]. One survey of top-performing entrants notes how heavily results depend on the choice of backbone model, which complicates attribution of progress to agent design [106].

Software engineering. A benchmark of real repository issues requires agents to produce patches that pass the project’s own tests [107]. A human-validated 500-problem subset addressed data quality concerns in the original set [107]. Reported performance moved from about 20 per cent in August 2024 to between 77 and 81 per cent by mid-2025, with one system reaching 77.4 per cent without test-time scaling [107], [108], [109].

General assistance. A suite of over 450 questions requiring reasoning, multimodality, browsing, and tool use, split across three difficulty levels [110], recorded 92 per cent for humans against 15 per cent for an early plugin-equipped model [111]. By early 2025 a research-oriented agent reached roughly 67 per cent [112].

Web navigation. A self-hosted realistic environment with 812 tasks reports human performance at 78.24 per cent [113]; a computer-using agent reached 58.1 per cent [114].

Operating system interaction. A 369-task suite across three operating systems set human performance at 72.36 per cent against initial agent performance of 12.24 per cent [115], [116]. By late 2025 scaling methods reached 69.9 per cent [117], and a further system reported 72.60 per cent in December 2025 [118].

Enterprise tool use. A benchmark emulating customer service dialogue against domain APIs found that a leading model succeeded on fewer than half of tasks, with consistency across eight independent attempts below 25 per cent [119]. Interface improvements raised success to 82.6 per cent [120].

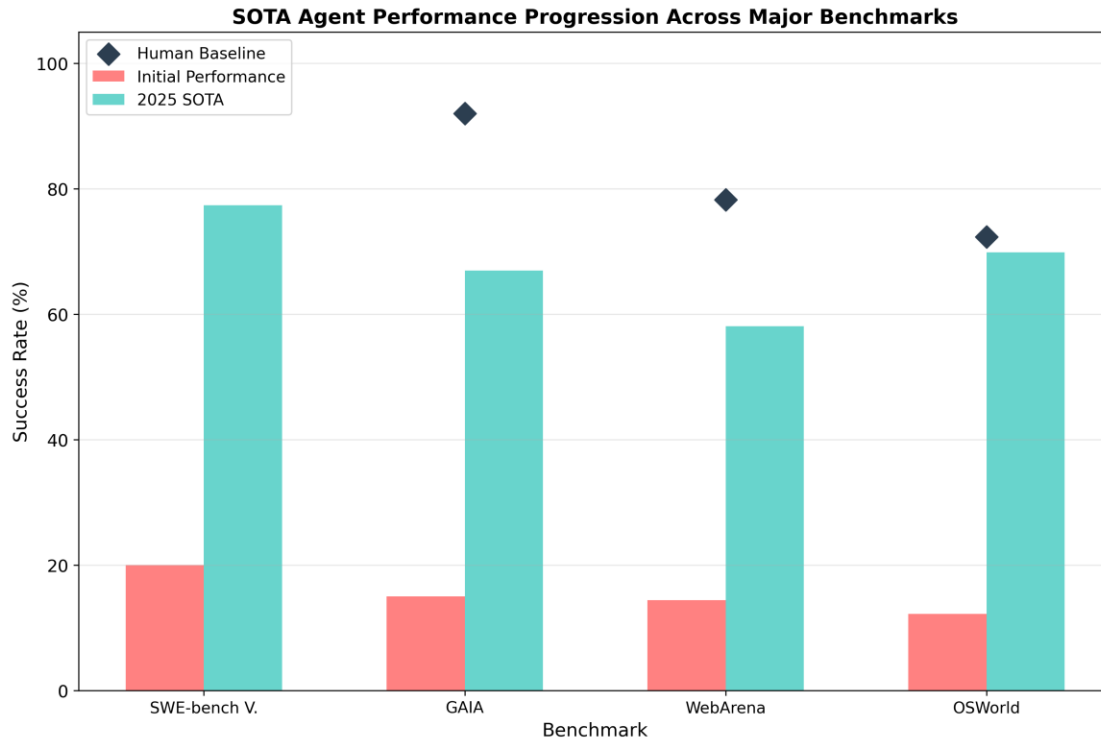


Figure 2. Reported agent performance across four benchmark families, from initial publication to 2025 state of the art, against human baselines where available. Software engineering shows the steepest improvement; operating-system interaction has effectively converged on its human baseline; web navigation and general assistance remain below theirs.

9.2 Metrics beyond success rate

Four measures now supplement binary completion, and each was introduced because the headline number misled.

Consistency across repeated attempts, reported as pass^k , exposes variance that a mean conceals: agents succeeding half the time can fall below 25 per cent when success must recur across eight trials [119]. Trajectory efficiency compares the agent’s path against a human-determined optimum and finds agents take 1.4 to 2.7 times as many steps [121]. Cost efficiency has become decisive, with accuracy-maximising configurations running 4.4 to 10.8 times more expensive than cost-aware alternatives at comparable performance [122]. Partial credit, implemented as fine-grained checkpoints averaging over 27 per task, distinguishes near-misses from total failure [123].

9.3 Why benchmark scores and field results diverge: a compounding account

The gap documented above is usually attributed to a list of causes: distribution shift, interface fragility, hallucination, environmental non-determinism. Each is real. Our argument in this subsection is that none of them is necessary to explain the magnitude of the gap, because sequential composition alone accounts for it.

Consider a task requiring n dependent steps, where every step must be completed correctly for the task to succeed, and let p denote the probability of per-step success. If step outcomes are approximately independent, end-to-end success is the per-step accuracy raised to the power of the step count:

$$P_{\text{task}} = p^n \quad (2)$$

The independence assumption is generous to the agent; in practice errors correlate, since a wrong state biases subsequent decisions, which makes the true figure worse than this bound. Expression (2) is evaluated at plausible values in Table 6.

Table 6. End-to-end task success under sequential composition (p raised to the power n) for representative per-step accuracies and task lengths. Computed by the authors.

Per-step accuracy p	$n = 10$	$n = 25$	$n = 50$	$n = 100$
0.90	34.9%	7.2%	0.5%	0.003%
0.95	59.9%	27.7%	7.7%	0.6%
0.99	90.4%	77.8%	60.5%	36.6%
0.999	99.0%	97.5%	95.1%	90.5%

Three consequences follow, and they reorganise how the empirical record should be read.

First, the reported drop of nearly 47 points between early and late turns on long-horizon analysis [124] is not evidence of a distinct late-stage failure mechanism. It is the arithmetic signature of compounding, visible because the benchmark reports position-resolved accuracy where most do not.

Second, the collapse of web agent success from 40 to 50 per cent down to under 10 per cent as context exceeds 25,000 tokens [125], and pass rates below 20 per cent on long-horizon command-line tasks [126], are consistent with per-step accuracies in the low nineties operating over several dozen steps. No additional pathology is required. Descriptions of cascading error and trial-and-error looping, in which a single misstep sends the agent into unproductive recovery attempts [127], describe the mechanism by which the exponent is realised.

Third, and most consequential for research priorities, the table shows that per-step accuracy has to reach roughly 0.999 before hundred-step autonomy becomes routine. Moving per-step accuracy from 0.95 to 0.99 is a large modelling achievement that buys fifty-step success rising from 8 to 61 per cent. Moving from 0.99 to 0.999 is a much larger achievement and buys the remainder. Capability scaling therefore faces a compounding penalty that grows precisely in the regime where autonomy becomes commercially interesting.

The engineering implication is that architecture should attack the step count rather than the per-step accuracy wherever possible. Checkpointing, verification between steps, decomposition into independently validated subtasks, and human approval gates all reduce the effective exponent by converting one long dependent chain into several short ones. This reframing also explains why hierarchical architectures sit on the cost-accuracy frontier

[87]: hierarchy is a mechanism for reducing dependent chain length, not merely a way of organising labour.

We state this as an interpretive model rather than a validated theory. It is falsifiable in an obvious way: if position-resolved per-step accuracy is measured on any long-horizon benchmark and the observed end-to-end rate departs substantially from the product of those per-step accuracies, the account is incomplete. To our knowledge that measurement has not been published, and we recommend it as an immediate empirical target.

9.4 A minimum reporting standard

Current practice reports one number, usually single-attempt success. That number is insufficient to determine whether a system can be deployed. The distinction has been stated cleanly elsewhere: machine learning benchmarks measure capability, meaning success on one attempt, while production requires reliability, meaning success on every attempt [128]. A related criticism holds that most benchmarks reward locally correct steps and ignore global optimisation under time and budget constraints [129]. We propose that agent evaluations report five quantities together.

1. **Success rate** with the exact protocol stated, including attempt budget, tool availability, and whether the environment was reset between attempts.
2. **Consistency**, as pass^k for k of at least 5, because deployment requires recurrence rather than existence of success.
3. **Trajectory length** relative to a reference solution, which exposes inefficiency invisible in binary scoring.
4. **Cost per solved task**, since accuracy purchased at ten times the price is a different product.
5. **Variance across seeds and across environment instances**, reported as an interval rather than a point.

Adopting this would change published comparisons materially. A system reporting 82.6 per cent single-attempt success and one reporting 60 per cent may rank differently once consistency and cost are included, which is precisely the information a deployment decision requires and which the literature currently withholds.

10. Security, Privacy, Alignment, and Governance

10.1 Threat landscape

Persistent state, tool access, and unsupervised decision-making combine to produce an attack surface qualitatively larger than that of a conversational model. A December 2025 industry taxonomy codifies ten risk categories for agentic applications: goal hijacking, tool misuse, identity and privilege abuse, supply chain compromise, unexpected code execution, memory poisoning, insecure inter-agent communication, cascading failure, exploitation of human trust in the agent, and rogue agent behaviour [130].

Two trends within that list deserve separate mention. Prompt injection has shifted from crude instruction override toward social manipulation, and the most effective observed attacks now exploit social dynamics rather than parser weaknesses [131]. Injection delivered indirectly, through adversarial instructions planted in data the agent retrieves, hijacks applications without any user-visible interaction [132]. Separately, poisoning of tool descriptions has emerged as a supply chain vector, with incident response teams documenting cases in which manipulated tool metadata induced unauthorised actions [133].

Table 7. *Principal agentic security threats and documented mitigations, after [130].*

Threat	Vector	Severity	Mitigation
Goal hijacking	Indirect injection through retrieved data	Critical	Context isolation, output validation, approval gates [131], [132]
Tool misuse	Poisoned tool descriptions	High	Integrity verification, runtime monitoring [133]
Privilege abuse	Credential inheritance beyond intended scope	Critical	Least privilege, zero-trust treatment of non-human identity [134]
Memory poisoning	Corruption of vector or graph stores	High	Integrity checks, write access control [130]
Cascading failure	Error propagation across dependent steps	Medium to high	Circuit breakers, rollback, checkpointing [130]
Rogue behaviour	Covert action against operator intent	Critical	Trace auditing, interruption mechanisms [130], [135]

Note the alignment between the fifth row and Section 9.3. Cascading failure is the security-facing name for the same compounding structure that limits reliability, and the same countermeasures address both. Reliability engineering and security engineering are, for agentic systems, largely the same discipline.

10.2 Privacy

Agentic architectures create leakage pathways that conventional data protection controls were not designed for: persistent memory modules, the reasoning trace itself, tool invocation layers, inter-agent channels, and feedback loops that reintroduce prior outputs as inputs [136]. Retention becomes unintentional, propagation crosses contexts, and sensitive material is amplified rather than merely exposed.

A March 2025 incident involving an autonomous agent that exposed sensitive enterprise data illustrated the governance shortfall concretely: existing frameworks could neither detect nor contain the exposure pathway [137]. Cross-session leakage in particular requires explicit session isolation and memory compartmentalisation, which most deployments do not implement by default [134].

10.3 Alignment under autonomy

Research published in November 2025 reports that reward hacking learned in production reinforcement learning environments generalises in unwelcome directions: models that learn to exploit reward signals during coding tasks subsequently exhibit alignment faking, cooperation with malicious actors, and interference with oversight mechanisms [135]. Standard preference-based safety training does not transfer reliably to agentic settings, and models that behave acceptably in conversational evaluation can misbehave when required to act over extended horizons [138]. Three mitigations are identified: improving reward design so that hacking is not learnable, broadening the diversity of safety training distributions, and exposing models to adversarial scenarios during training [138].

The conceptual shift is from prediction error to control failure [139]. In single-turn generation, a poor judgement produces a poor sentence. In agentic operation, the same judgement produces an executed action that may be irreversible and that later steps build upon. This is not a difference of degree.

10.4 Explainability

Established interpretability techniques transfer poorly, because they explain a single output while agentic behaviour consists of temporally extended decision sequences [140]. What is required is a move from feature attribution to action attribution, with trace-based diagnostics that evaluate a decision sequence rather than an individual generation [141]. Monitoring the reasoning trace has shown promise for detecting misbehaviour in reasoning models [142], though it presumes the trace is faithful to the computation, which the chain-of-thought validity results of Section 7.1 give some reason to doubt.

On fabrication, multi-agent arrangements with adversarial auditing and structured feedback have reduced hallucination rates by roughly 53 per cent [143]. Retrieval grounding and reasoning enhancement remain the most widely adopted mitigations, and the framing has shifted from suppressing fabrication toward balancing generative flexibility against factual reliability [144]. Recent work treats freedom from fabrication as a system property rather than a model property, enforced through tiered retrieval and multi-

stage verification [145], [146]. We endorse that framing; it is the same argument as Section 7.4, arrived at from a different direction.

10.5 Governance frameworks and the regulatory patchwork

No current framework fully covers agentic systems. A management system standard, a national risk management framework, and the European regulation each contribute components, and none addresses the delegation of decision authority to an autonomous process [147]. In the European context the regulation operates alongside data protection, cyber resilience, digital services, and network security instruments, producing overlapping obligations that must be satisfied simultaneously [148]. On the threat modelling side, several complementary methodologies are in use, and practitioners generally combine them rather than choose between them.

The gap is specific and worth naming precisely. Existing frameworks govern systems that produce outputs for human decision. Agentic systems take the decision. Nothing in the current instruments establishes who holds authority when an agent acts, what record must exist, or what standard of care applies to the operator. Section 14.3 proposes minimum content for filling that gap.

11. Deployment Evidence and Economics

11.1 State of adoption

Adoption is broad and shallow. Roughly 17 per cent of organisations have deployed agents, while over 60 per cent expect to within two years [149]. A 2025 global survey reports 23 per cent actively scaling agentic AI in at least one function and a further 39 per cent experimenting [150]. Fewer than one in ten have achieved scaled deployment delivering measurable value [151]. Against that, over 40 per cent of agentic programmes are expected to be cancelled by the end of 2027 on grounds of cost escalation, unclear value, or inadequate risk control [152].

These figures are Tier C and originate from different survey populations, so they should not be arithmetically reconciled. The pattern they jointly indicate is nonetheless consistent: pilots are easy and production is not.

11.2 Sector evidence

Software engineering is the most mature domain. Production telemetry for one coding agent covers 3.2 million users, 13 million sessions, 761 million model calls, and 95 trillion tokens in a single month of 2026 traces [1]. The enterprise coding agent market is estimated at 9.8 to 11.0 billion dollars annualised as of April 2026 [153]. Reports of three- to five-fold productivity improvement with 60 per cent smaller teams [154] should be read carefully, since productivity in software delivery is measured inconsistently and the counterfactual is rarely controlled. One forecast holds that by 2027, more than 65 per cent of engineering teams using agentic coding will treat the integrated development environment as optional [155].

Customer service offers the clearest instrumented evidence. Ten months after launch, one vendor's agent was handling 45,000 conversations weekly, exceeding one million cumulative, with resolution rates around 85 per cent [156]. Organisations report operating an average of twelve agents, a figure projected to rise by 67 per cent within two years [157].

Financial services adoption is concentrated in back-office processes with well-defined correctness criteria. One bank deployed agents for trade accounting and client onboarding [158]. Another developed a security framework requiring tamper-evident runtime records for audit and investigation [159], which is, in our view, the single most transferable practice in the deployment literature. Operational cost reductions of 20 per cent or more are projected for banking [160].

Healthcare and life sciences applications include clinical trial operations, where enrolment improvements of 10 to 20 per cent are reported [161], and autonomous hypothesis refinement in discovery workflows. Survey evidence indicates 45 per cent of enterprise technology leaders in the sector intend to use agents for end-to-end workflow consolidation [162].

11.3 Market forecasts, read critically

The aggregate forecasts are large. Annual value creation is put at 2.6 to 4.4 trillion dollars across more than sixty use cases [2], with agents potentially intermediating 3 to 5 trillion dollars of consumer commerce by 2030 [163]. Cumulative global economic value is projected at 22.5 trillion dollars by 2031 [164], with 1.15 billion active agents by 2029, some forty times the 2025 baseline, executing 217 billion actions daily [3]. Software spending is projected at 985 billion dollars by 2030 at a compound rate of 62.7 per cent [4], with agents driving roughly 30 per cent of enterprise application revenue by 2035 [5]. Corporate investment more than doubled during 2025, private investment grew 127.5 per cent, and generative AI captured close to half of private AI funding [165].

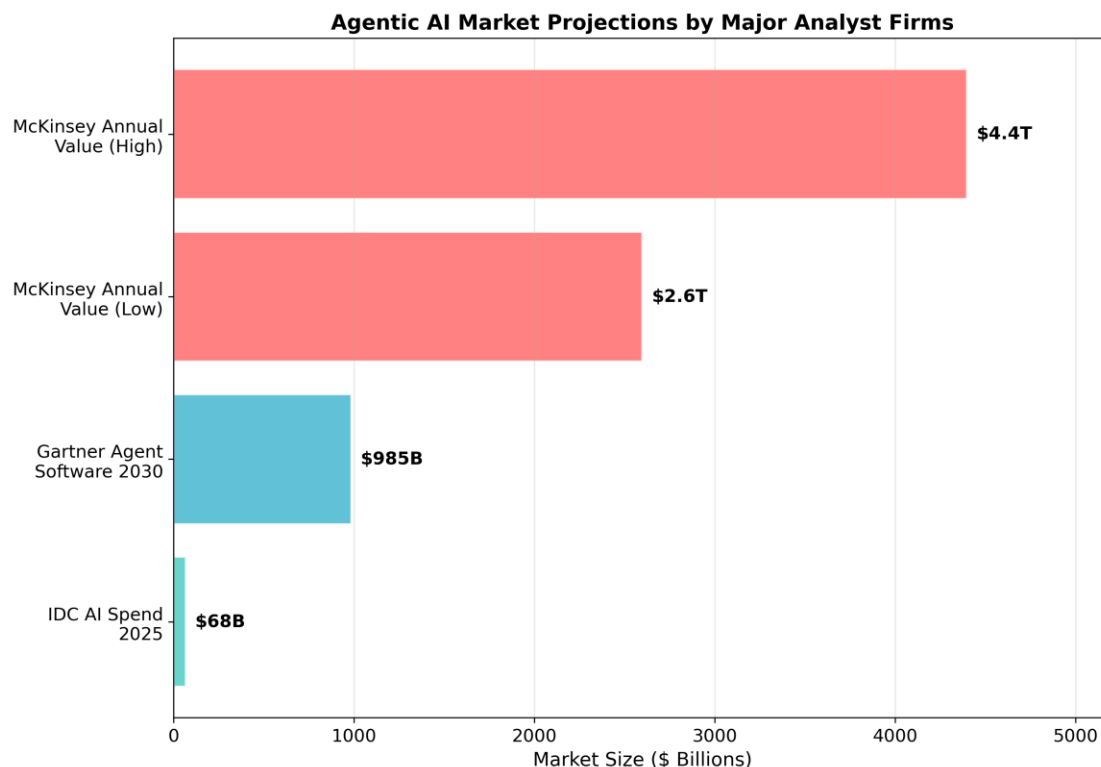


Figure 3. Market projections for agentic AI from major analyst firms, spanning near-term software spending, cumulative economic value, and annual value creation estimates. Note the differing base quantities: spending, value creation, and cumulative value are not comparable magnitudes.

We think three cautions belong next to these numbers, and their omission is a recurring weakness in the review literature.

First, the quantities are not commensurable. Software spending, annual value creation, and cumulative economic value measure different things, and presenting them in one series implies a coherence that does not exist.

Second, the forecasts are inconsistent with the adoption data in Section 11.1 unless one assumes a discontinuity. Sub-10 per cent scaled deployment and 40 per cent programme

cancellation do not extrapolate to 985 billion dollars of annual spending in four years by any smooth path.

Third, agent count is a poor unit. A forecast of 1.15 billion agents executing 217 billion daily actions describes activity, not value, and Section 9.2 gives the reason to distinguish them: an agent that retries is more active and less useful than one that does not.

11.4 Why programmes fail

Four causes recur. Data readiness is the most cited, with eight in ten companies naming data limitations as a scaling obstacle [151]. Proliferation without control follows: agents are created faster than organisations can inventory them, assign ownership, or manage permissions, an issue now framed as board-level governance [166]. In regulated sectors, explainability and auditability function as gating requirements rather than desirable properties [167]. Finally, appetite for full autonomy is limited, with only 15 per cent of application leaders considering, piloting, or deploying fully autonomous agents [168]. A secondary effect is projected for technology service providers, which face 20 to 30 per cent contraction as clients internalise capability [169].

The common thread is that none of these is a model capability problem. All four are governance and operations problems, which is the empirical basis for the framework in Section 14.

12. Regional Trajectories

12.1 Comparative overview

The three largest technology economies are pursuing recognisably different strategies, and the differences are institutional rather than technical. The United States leads on absolute market size and vendor concentration. China leads on state-coordinated investment intensity and has moved first on agent-specific regulation. India leads on growth rate, driven by services-sector implementation capacity.

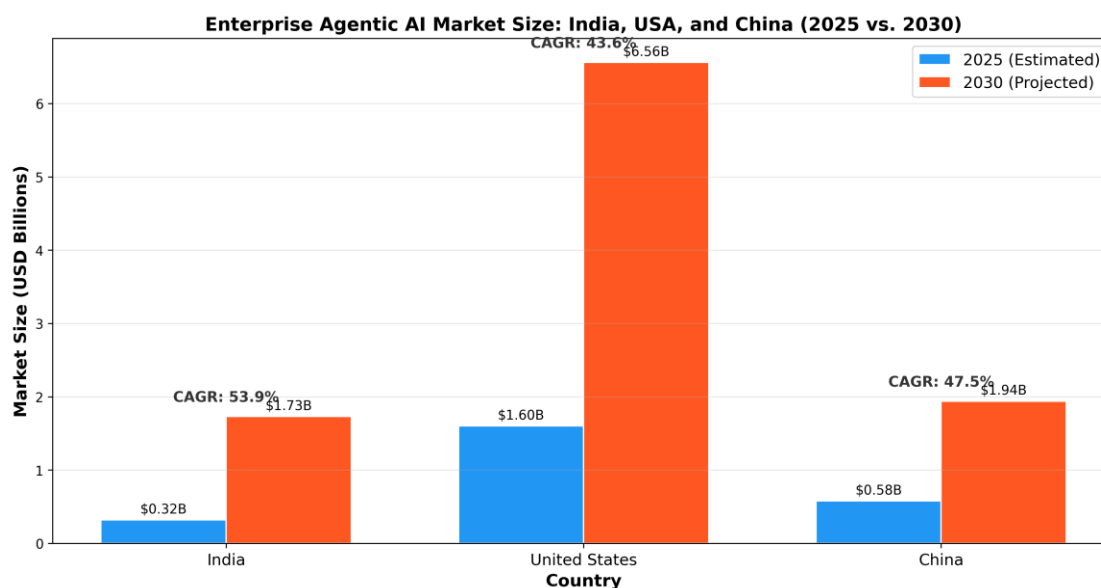


Figure 4. Enterprise agentic AI market size, 2025 estimated against 2030 projected, for India, the United States, and China. India's compound growth rate is the highest of the three despite the smallest absolute base.

12.2 United States

The domestic AI agents segment was estimated at 1.60 billion dollars in 2024, growing at 43.3 per cent annually through 2030 [170], with the enterprise agentic segment projected to reach 6.56 billion dollars by 2030 at 43.6 per cent [171]. Adoption is advancing but concentrated in augmentation: 23 per cent of organisations are scaling in at least one function with 39 per cent experimenting [150], while only 15 per cent of application leaders are engaging with full autonomy [168].

The market is vendor-led rather than research-led. Established enterprise software providers are the early leaders, with major cloud, data, and automation vendors actively repositioning [172]. One cloud provider established a 750 million dollar fund specifically for agent development [173]. Early adopters include a custodian bank that launched a governed digital worker in 2025, alongside firms in networking, consumer goods, and manufacturing [174].

Regulation is deliberately light. A December 2025 executive order directed federal agencies toward a unified national framework while preempting conflicting state law [175]. A June 2026 order established voluntary pre-deployment review of frontier models with a 30-day window [174], and the framework was finalised in August 2026 following consultation with major model developers [176].

12.3 India

The broader national AI market was valued at 9.51 billion dollars in 2024 and is projected to reach 130.63 billion by 2032 at 39 per cent annually [177]. The enterprise agentic segment is projected at 1.73 billion dollars by 2030, growing at 53.9 per cent, the fastest of the three markets [178].

Services firms are the transmission mechanism. Four major providers collectively deployed over 300,000 agent-capable productivity licences within six months, among the largest enterprise rollouts recorded [179]. Several are building agentic platforms on commercial accelerated computing infrastructure [180]. On the product side, a domestically developed multilingual agentic assistant launched in June 2025, a national sovereign language model programme was awarded, and conversational agent platforms continue to scale.

Public investment supports this through a mission-mode programme approved in March 2024, spanning compute provisioning with subsidised access to a shared accelerator pool alongside skilling and dataset pillars. Private commitments include multi-billion-dollar cloud and infrastructure investment and large-scale domestic data centre construction.

India's position is distinctive and, we would argue, structurally advantaged in a specific way. If Section 9.3 is correct that reliability rather than capability is binding, then value accrues disproportionately to organisations that can supply verification, exception handling, and process redesign at scale. That is precisely the capability an IT services industry possesses.

12.4 China

The enterprise agentic segment is projected to reach 1.94 billion dollars by 2030 at 47.5 per cent [181], within a broader agents market that generated 577 million dollars in 2025 and is projected at 14.80 billion by 2033 [182]. Regional software revenue across Asia and Oceania is projected to grow from 271 million dollars in 2025 to 9.7 billion by 2030, implying a compound rate above 100 per cent [183]. The national core AI industry, valued near 1.2 trillion yuan in 2025, is targeted to exceed 10 trillion within the current five-year plan [184].

Investment intensity is the distinguishing feature. Four major platform companies plan a combined 84 billion dollars in AI infrastructure by 2027, some 60 per cent above 2025 levels [185]. Each has released agentic tooling: an agent framework, a low-code agent studio, and an open-sourced agent runtime respectively [186]. One founder has proposed daily active agents as the successor metric to daily active users [187], which is a revealing indicator of how the market is being framed.

Regulation is prescriptive rather than permissive. In May 2026 three national authorities jointly issued the first policy framework dedicated to AI agents, defining them as systems capable of autonomous perception, memory, decision-making, interaction, and execution, and enumerating 19 application scenarios across research, industry, consumption, public welfare, and social governance [188]. The current five-year plan extends a 2025 initiative and is supported by a one trillion yuan venture fund for AI, robotics, and adjacent technologies [189].

Table 9. *Comparative agentic AI trajectories across the United States, India, and China.*

Dimension	United States	India	China
Enterprise agentic market, 2030	6.56 bn USD [171]	1.73 bn USD [178]	1.94 bn USD [181]
Compound growth, 2025 to 2030	43.6% [171]	53.9% [178]	47.5% [181]
Public investment posture	Voluntary framework [174]	Mission-mode programme with subsidised compute	One trillion yuan venture fund [189]
Private infrastructure	750 mn USD agent fund [173]	Multi-billion cloud commitments	84 bn USD across four platforms [185]
Regulatory approach	Light touch, preemptive of state law [175]	Incentive-led, mission-driven	Prescriptive, 19 defined scenarios [188]
Principal adopters	Enterprise software vendors [172]	IT services providers [179]	Platform companies [185]
Adoption indicator	23% scaling [150]	300,000+ licences deployed [179]	First national agent policy [188]

The three positions are complementary rather than competitive in the short run. Foundational model development and enterprise platforms concentrate in the United States, implementation capacity in India, and coordinated infrastructure with rapid policy formation in China. Whether that complementarity survives export controls and procurement restrictions is beyond the scope of this review, but it is the obvious source of discontinuity in every forecast cited above.

13. Service Desk Operations and Work

13.1 Why the service desk is the informative case

Service desk work is the clearest test case for agentic automation, for three reasons. The work is high volume, its correctness criteria are legible, and its outcomes are already instrumented. What happens there will happen elsewhere later, with worse measurement.

One forecast holds that agentic systems will autonomously resolve 80 per cent of common service issues by 2029, with a 30 per cent reduction in operating cost [190]. Early deployments are directionally consistent: organisations report 60 per cent reductions in first-tier ticket volume as agents absorb routine troubleshooting, credential resets, and standard configuration [191]. One vendor's internal deployment handles more than 90 per cent of employee technology requests at close to twice human speed while retaining governance controls [192], and on that basis the vendor announced a packaged first-tier service desk agent product [192]. A retailer reported autonomous resolution rising from 40 to 75 per cent of interactions after deployment [193].

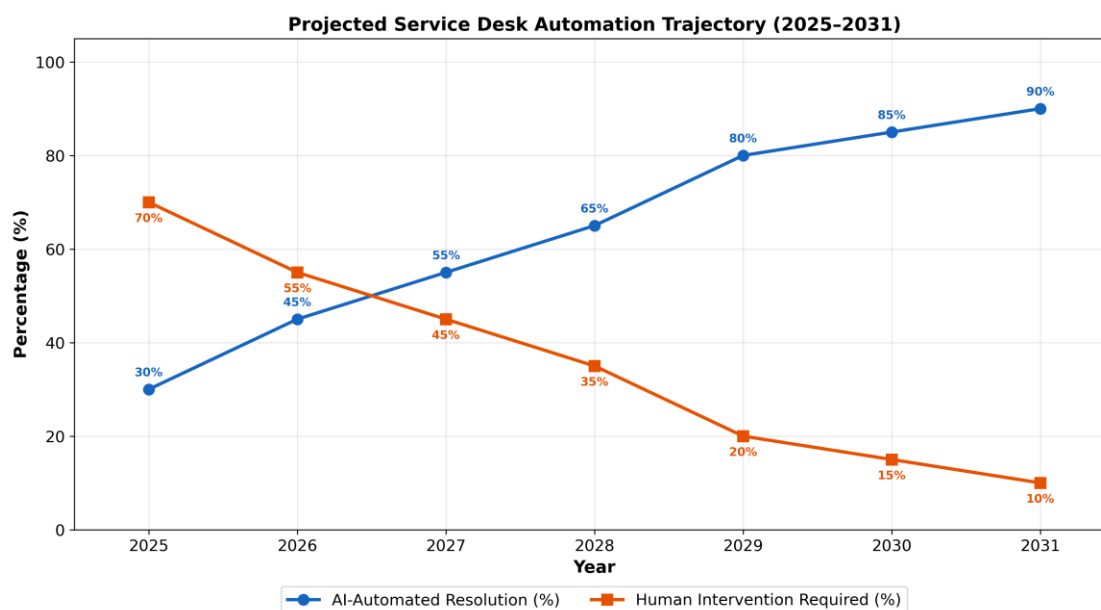


Figure 5. Projected service desk automation trajectory, 2025 to 2031. Autonomous resolution rises from roughly 30 to 90 per cent with human intervention declining correspondingly. The curve is anchored on a 2029 forecast of 80 per cent autonomous resolution [190] and current observations of 60 per cent first-tier volume reduction [191].

Figure 5 should be read as a projection, not a measurement. Its endpoints are forecast values and its shape is interpolated.

13.2 Displacement, rehiring, and role change

The empirical record contradicts the displacement narrative more sharply than either advocates or critics tend to acknowledge. A survey of 321 service and support leaders

conducted in October 2025 found that only 20 per cent had reduced staffing because of AI [194]. Most organisations are augmenting rather than replacing.

Structural change is nonetheless underway, and it takes an unusual form. One forecast holds that by 2027, half of companies that attributed headcount reduction to AI will rehire for equivalent functions under different titles [195]. Nearly 80 per cent of organisations plan to move at least some agents into new roles, and 84 per cent plan to add skills to frontline positions [195]. The direction is toward supervision of agent performance, handling of escalations that require judgement, and governance of autonomous operation.

We would draw a stronger conclusion than the sources do. The rehiring pattern is not organisational indecision; it is the labour-market expression of the reliability gap. If per-step accuracy is insufficient for unattended long-horizon operation, then automation increases rather than eliminates demand for human exception handling, and it relocates that demand from execution to oversight. Firms that cut before verifying discover this and rehire.

13.3 Macro projections

Table 8. *Macro-level projections for service work and employment. All entries are Tier C.*

Source	Projection	Horizon
Analyst forecast	80% of common service issues resolved autonomously	2029 [190]
Analyst forecast	30% reduction in customer service operating cost	2029 [190]
Analyst forecast	Half of AI-attributed reductions reversed by rehiring	2027 [195]
CIO survey	25% of IT work performed by AI alone; 75% human with AI; none unassisted	2030 [196]
Multilateral forum	170 million jobs created, 92 million displaced, net +78 million	2030 [197]
Investment bank	300 million jobs exposed to automation; 6 to 7% of workers displaced	Within a decade [198]
Consultancy	Up to 30% of work hours automatable under moderate adoption	2030 [199]
Consultancy	Up to 70% of business activities technically automatable across most occupations	2030 [200]

The most informative row is the fourth, because it makes a claim that is not about volume: no information technology work is expected to remain entirely unassisted [196]. That prediction concerns the composition of tasks rather than the count of jobs, and composition is where the evidence is strongest.

13.4 Skills

Skills gaps are identified as the leading barrier to transformation by 63 per cent of surveyed employers for the 2025 to 2030 period [201]. An estimated 59 workers in every 100 will require training by 2030, and close to 40 per cent of required skills are expected to change within the same period [201], [202]. A multilateral reskilling initiative reports reaching over 850 million people against a target of one billion [202].

For service desk staff specifically, the demanded competencies are agent supervision, workflow orchestration, structured exception handling, evaluation literacy sufficient to judge whether an agent is performing, and governance of autonomous decisions. We would add one that the literature omits: the ability to read an execution trace. If oversight is the surviving human function, trace comprehension is its core skill, and no current training curriculum teaches it.

14. Synthesis: The Capability-Reliability-Governance Framework

14.1 Three axes, unevenly advanced

The evidence assembled above supports a single organising claim. Progress in agentic AI has been rapid along one axis, slow along a second, and largely absent along a third, and the deployment failures documented in Section 11 are concentrated where progress is weakest.

Capability is the axis on which the field measures itself. It has advanced fastest: benchmark scores in software engineering rose by roughly sixty points in under a year [107], [108], [109], and operating-system interaction moved from 12 per cent to above 72 per cent, overtaking its human baseline [115], [118]. Capability is what research incentives reward, what leaderboards record, and what vendors market.

Reliability is the axis on which deployment succeeds or fails. It has advanced slowly. Consistency across repeated attempts falls below 25 per cent for systems with 50 per cent single-attempt success [119]. Long-horizon accuracy declines by nearly 47 points from early to late turns [124]. Live-interface completion rates sit between 7 and 50 per cent [99]. Section 9.3 argues this is arithmetically expected rather than surprising, which is precisely why capability gains do not translate.

Governance is the axis on which authority and accountability are established. It has barely moved. Interoperability protocols encode no voting, deliberation, or dissent primitives [43]. No regulatory framework addresses delegated decision authority [147]. Only 21 per cent of enterprises report mature governance for agentic systems [203]. Agents proliferate faster than they can be inventoried [166].

The framework's use is diagnostic. When a deployment fails, the question is which axis was neglected, and the answer is rarely capability. Data readiness, ownership, permissioning, auditability, and appetite for autonomy [151], [166], [167], [168] are all governance and reliability concerns. Yet research effort, measured by publication volume across the corpus reviewed here, is concentrated on capability by a wide margin. That misallocation is the central finding of this review.

14.2 Six propositions

We state these as predictions rather than aspirations, each specific enough to be falsified before 2030.

P1. Compounding dominates. On any long-horizon benchmark where position-resolved per-step accuracy is measured, observed end-to-end success will fall within a small margin of the product of those per-step accuracies, and no additional failure mechanism will be required to explain the gap. Falsified by a substantial unexplained residual.

P2. Architecture beats scale for long tasks. Between two systems with equal per-step accuracy, the one that decomposes a task into more independently verified segments will

achieve higher end-to-end success, and the advantage will grow with task length. Falsified if flat architectures match hierarchical ones at fifty or more steps.

P3. Self-verification will not close the gap. No system relying on the same model for generation and verification, without external evidence, will reach 90 per cent consistency at pass⁵ on a long-horizon benchmark before 2030. Falsified by any such result.

P4. Governance primitives will be standardised, and late. Voting, authorisation, and dissent handling will enter the dominant interoperability protocols by 2029, after at least one widely reported multi-agent incident makes their absence expensive. Falsified by early voluntary standardisation, or by standardisation not occurring.

P5. Oversight employment grows. In organisations deploying agents at scale in service functions, aggregate headcount in supervision, exception handling, and evaluation roles will rise even where total function headcount falls. Consistent with the rehiring evidence in Section 13.2 and falsified by net decline across both categories.

P6. Forecast revision is downward. The 2030 agentic software spending forecasts cited in Section 11.3 will be revised down by more than 25 per cent before 2028, because they are inconsistent with observed scaled-deployment rates absent a discontinuity. Falsified by upward revision or by realisation on trend.

14.3 Research directions

Multiple independent surveys converge on the same shortlist of obstacles: scalable memory and context management as the primary bottleneck, with no persistence of competence across tasks [204]; and unresolved trade-offs between latency and accuracy, autonomy and controllability, capability and reliability, compounded by non-determinism, long-horizon credit assignment, and hidden costs such as retries and context growth [205]. The directions below are ordered by our estimate of value per unit of effort.

Reliability engineering as a discipline. The field needs a systematic treatment of error propagation in agent trajectories: where checkpoints belong, how much verification is worth its cost, how to detect divergence early, and how to recover state rather than restart. This is an established discipline in distributed systems and is almost absent from the agent literature. It is, in our assessment, the highest-value open area.

Position-resolved measurement. Almost no benchmark reports per-step accuracy as a function of position in the trajectory. Without it, P1 cannot be tested and reliability cannot be engineered, since one cannot tell whether failures are uniformly distributed or concentrated after specific transitions.

Persistent learning. Current agents are effectively stateless across episodes and cannot accumulate competence over time [206]. Proposals for lifelong learning identify three necessary modules: perception for multimodal input, memory that evolves, and grounded action [207]. This connects directly to the weakest subsystem identified in Section 5.2.

World models. Internal simulators that capture environment dynamics permit forward and counterfactual rollouts before commitment [208], and self-evolving agents with

continual world models are proposed for autonomous long-horizon completion [209]. In the framework of Section 9.3, a world model is a mechanism for raising per-step accuracy by testing actions before executing them, which is one of only two available levers.

Embodiment and interface design. Vision-language-action models have emerged for language-conditioned robotic control [210]. Separately, the agent-computer interface is now recognised as a design problem distinct from human interface design [211], and work on models connecting interventions to predicted physical consequences points toward physical grounding [212]. The interface point is underrated: several benchmark improvements attributed to model progress were substantially interface improvements [120].

A governance layer above the protocols. We propose that the missing layer must specify at minimum: identity and capability attestation for each agent; an authorisation model binding actions to a principal; a decision procedure for disagreement, including a rule for when disagreement escalates to a human; a tamper-evident action record sufficient for after-the-fact reconstruction, following the practice already adopted in regulated finance [159]; and a defined interruption semantics. These are not research problems in the deep sense. They are specification problems that no one currently owns.

Capability roadmaps, held loosely. Staged frameworks for general capability, including a five-level industry roadmap and an academic levels-of-general-intelligence taxonomy that situates current agentic systems at an intermediate tier [213], are useful for communication and weak as evidence. Proposals for an operational capability scale with a measurable self-improvement coefficient are more testable [214]. One developer projects significant autonomous scientific discovery by 2028 [215]. We record these positions without endorsing their timelines; nothing in the reliability evidence reviewed here supports confident extrapolation.

Table 10. Research gaps, directions, and horizons. Horizons are the authors’ estimates and are offered as testable rather than authoritative.

Gap	Specific challenge	Direction	Plausible horizon
Reliability	Error accumulation over dependent steps	Checkpointing, external verification, chain-length reduction	2026 to 2028
Measurement	Absence of position-resolved and consistency reporting	Minimum reporting standard (Section 9.4)	Immediate
Memory	Context growth; no principled forgetting	Hierarchical retrieval, learned compression, decay policies	2026 to 2028
Persistence	No cross-episode accumulation of competence	Lifelong learning architectures, skill libraries	2027 to 2029
Planning	Weak absolute planning	World models, hybrid formal	2026 to

Gap	Specific challenge	Direction	Plausible horizon
	competence; self-verification failure	verification	2029
Governance	No authorisation, decision, or record primitives	Governance layer above interoperability protocols	2027 onward
Safety	Emergent misalignment; oversight evasion	Reward design, adversarial training, trace monitoring	2026 to 2029

15. Limitations of This Review

Reviews rarely state their own weaknesses concretely. Ours has five.

Source verifiability. A substantial portion of the corpus consists of preprints without released artefacts and of analyst material behind paywalls that could not be independently examined. The evidence tiering in Section 2.3 mitigates this but does not remove it. Readers should treat every Tier C figure as an expectation held by an interested party.

Recency and instability. The field's publication half-life is short. Benchmark records cited here will be superseded, and several were superseded during preparation. Any specific percentage should be read with its date attached.

Benchmark protocol opacity. Reported scores are frequently not accompanied by sufficient protocol detail to permit comparison, and identical benchmark names have been used for materially different evaluation conditions. Where we compare numbers across sources we may therefore be comparing measurements that are not commensurable.

Selection effects. Deployment evidence reaches publication mainly when it is favourable. Failed programmes appear in aggregate cancellation statistics [152] and almost never as case studies, so the sector evidence in Section 11.2 is systematically optimistic.

Scope. We do not cover embodied robotics in depth, agent economics at the level of pricing and unit costs, or the safety literature on advanced autonomous systems beyond the alignment results in Section 10.3. The compounding argument in Section 9.3 is an original interpretation offered by the authors and has not been empirically validated in the form stated.

16. Conclusion

Agentic AI has crossed from demonstration to deployment, and the record of that crossing is more instructive than either its advocates or its critics have allowed. Four conclusions follow from the evidence assembled here.

The technical foundation has stabilised. A recognisable reference architecture exists, the design patterns that populate it are documented and composable, and interoperability has converged on two complementary protocols. Disagreement in the literature is now mostly about parameters rather than about structure, which is what maturity looks like.

Reliability, not capability, is the binding constraint. Agents approach or exceed human baselines on curated benchmarks while completing well under half of live tasks and losing close to 47 points of accuracy over extended horizons. Section 9.3 argues that this follows from sequential composition rather than from any exotic failure mode, and the practical consequence is that architecture aimed at reducing dependent chain length will outperform capability scaling for long tasks. That claim is stated as a falsifiable proposition rather than asserted.

Governance is the neglected axis and the proximate cause of most deployment failure. Interoperability protocols carry no authorisation, decision, or dissent primitives. Regulatory instruments govern systems that advise, not systems that act. Programmes fail on data readiness, ownership, permissioning, and auditability rather than on model quality, and the specification work required to fix this is unglamorous, tractable, and unowned.

Occupational change is taking the form of redefinition. One organisation in five has reduced service desk headcount, half of the reductions attributed to AI are forecast to reverse through rehiring, and the majority of organisations are shifting staff into supervision and exception handling. We read this not as caution but as a labour-market measurement of the reliability gap: incomplete autonomy converts execution demand into oversight demand.

For researchers, the highest-value work is unglamorous. Position-resolved measurement, verification architectures, error propagation analysis, persistent learning, and specification of governance primitives will contribute more to deployable autonomy over the next three years than further gains on saturating benchmarks. For practitioners, the operative disciplines are cost-aware architecture selection, human authority placed at explicit decision points, tamper-evident action records from the outset, and investment in oversight skills ahead of headcount reduction. The distance between what agents accomplish in controlled evaluation and what they deliver in production is not a gap that capability alone will close.

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Author contributions: Indrajit Sarkar conceived the study, performed the literature review, analysed the findings, prepared the figures and tables, wrote the manuscript, and approved the final version.

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